CLINICAL · APPROACH TO

Approach to postnasal drip and catarrh

Postnasal drip, or catarrh, is the feeling of mucus running down the back of the nose and throat, with the urge to clear the throat and sometimes a cough. It is one of the most common and most frustrating complaints in rhinology, because the sensation often outlasts any visible mucus. The approach is to find and treat what can be treated, the nose, the sinuses and reflux, to recognise the patients whose problem is a hypersensitive throat rather than excess mucus, and to make sure that nothing serious in the nasopharynx, larynx or chest has been missed.

AT A GLANCE

- Endoscope the nose and nasopharynx.
- Treat rhinosinusitis first: saline and steroid spray for 2 to 3 months.
- Catarrh without findings: explain, avoid antibiotics.
- Persistent unilateral signs: biopsy.

1 CHAPTER ONE What the patient means

Mucus, sensation or cough.

1.1 Definition and mechanism

Normally, a litre or so of mucus a day is carried backwards by the cilia and swallowed without being noticed. A sensation of postnasal drip arises when the volume rises, when the mucus becomes thicker or more irritant, when mucociliary clearance slows, or when the sensory nerves of the pharynx become hypersensitive. Many patients have more than one of these. In chest medicine, a cough attributed to the nose is called *upper airway cough syndrome*, but many patients with postnasal drip have no cough, and many with chronic cough have a sensitised cough reflex rather than a nasal cause.

DO NOT MISS

- **Unilateral nasal symptoms, blood-stained mucus or a neck lump:** nasopharyngeal and sinonasal tumour.

- **Hoarseness, dysphagia, weight loss or pain on swallowing:** laryngeal or pharyngeal malignancy.
- **A smoker with a cough that has changed, haemoptysis or breathlessness:** chest X-ray and respiratory referral.
- **Persistent cough in a patient taking an ACE inhibitor:** the drug.

2 CHAPTER TWO History

Four sources to explore.

2.1 The questions

Ask what exactly is felt: mucus that can be spat out, or a sensation that nothing relieves; and whether the mucus is clear, thick or discoloured. Ask about the nose, blockage, discharge, sneezing, smell loss and facial pressure, which point to rhinitis or rhinosinusitis. Ask about heartburn, a sour taste, hoarseness worse in the morning, globus and throat clearing, which point to reflux. Ask about cough, its duration (chronic cough lasts eight weeks or more), smoking, asthma and medication, including ACE inhibitors and drugs that dry the mucosa (antidepressants, anxiolytics, anticholinergics). The *Reflux Symptom Index* and *SNOT-22* quantify the symptoms.

3 CHAPTER THREE Examination

Look at both ends.

3.1 Nose, nasopharynx and larynx

Examine the nose with an endoscope, looking for mucopurulent discharge in the middle meatus or sphenoethmoidal recess, polyps, adenoidal tissue, a Thornwaldt cyst and any nasopharyngeal mass. Pass the scope down to the larynx and look for signs associated with reflux (posterior commissure thickening, oedema, erythema) and for any lesion that explains the throat symptoms. Examine the neck. A CT of the

sinuses is arranged only if chronic rhinosinusitis is suspected and not settling on medical treatment.

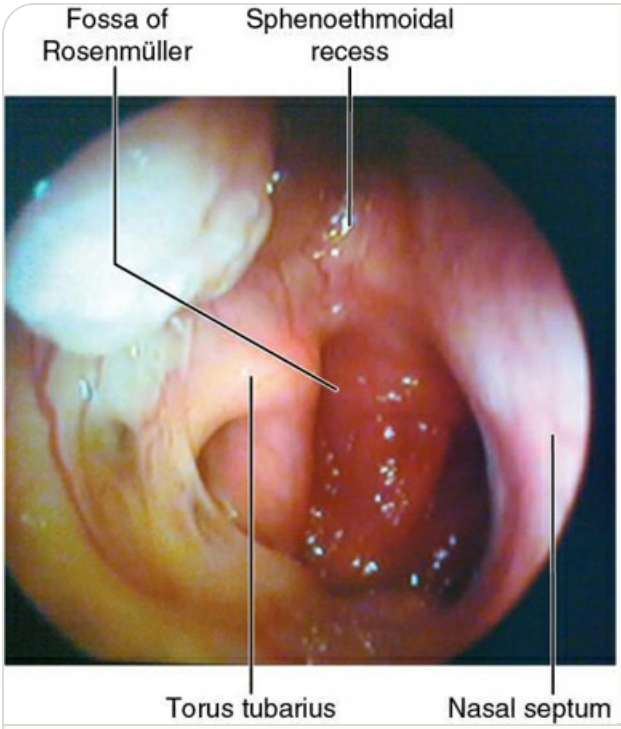


FIGURE 1 The region of the Eustachian tube, sphenoethmoidal recess, choana and fossa of Rosenmüller, examined at endoscopy.
Dhingra, fig. 89.1

4 CHAPTER FOUR Causes

Where the drip comes from.

CAUSES OF POSTNASAL DRIP

SOURCE	EXAMPLES	CLUE
Nose	Allergic and non-allergic rhinitis, rhinitis of the elderly	Sneezing, watery discharge, triggers
Sinuses	Chronic rhinosinusitis, dental sinusitis, fungal ball	Pus in the meatus, smell loss, unilateral
Nasopharynx	Adenoiditis, Thornwaldt cyst, tumour	Endoscopic findings
Clearance	Primary ciliary dyskinesia, cystic fibrosis, dry atrophic nose, after sinus surgery	Thick mucus, crusting, lifelong symptoms

SOURCE	EXAMPLES	CLUE
Throat and reflux	Laryngopharyngeal reflux, globus	Throat clearing, hoarseness, normal nose
Sensory	Cough hypersensitivity, laryngeal hypersensitivity	Cough triggered by talking, cold air, scents
Drugs	ACE inhibitors, drying drugs	Medication history

5 CHAPTER FIVE Management

Treat what is found.

5.1 When a cause is found

Rhinitis is treated with saline irrigation and a steroid spray, with an antihistamine for allergy and ipratropium for watery discharge. **Chronic rhinosinusitis** follows its own pathway: saline, steroids, a long course of a macrolide in selected patients, and surgery if medical treatment fails. **Reflux** symptoms are managed with diet and lifestyle changes, alginates and, for typical acid symptoms, a proton pump inhibitor trial; evidence for proton pump inhibitors in throat symptoms alone is weak. Any drying or causative medication is reviewed with the prescriber.

5.2 When the sensation persists

Many patients have a normal nose and larynx and still feel the drip. Explanation, reassurance that nothing serious has been found, adequate hydration and steam inhalation help. Throat clearing perpetuates the irritation, so **speech and language therapy** for laryngeal hygiene and cough suppression techniques is effective. In refractory chronic cough, a neuromodulator such as low-dose amitriptyline or **gabapentin** may be tried under specialist care.

POSTNASAL DRIP

1 Red flags?

→ unilateral symptoms, blood, neck lump, hoarseness or dysphagia: full endoscopic examination and imaging

→ **smoker with changed cough**
chest X-ray

2 Nasal endoscopy

→ **rhinitis**
saline and steroid spray, ipratropium if watery

→ **pus or polyps**
chronic rhinosinusitis pathway

→ **nasopharyngeal lesion**
biopsy

3 Throat and larynx

→ **reflux features**
lifestyle, alginate, trial of proton pump inhibitor

→ **normal**
step 4

4 Normal nose and larynx

→ review drugs

→ explanation and hydration

→ speech therapy for throat clearing

→ neuromodulator for refractory cough

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The mechanisms:** volume, viscosity, clearance, hypersensitivity.
- **Upper airway cough syndrome** and cough hypersensitivity.
- **The red flags** that need endoscopy and imaging.
- **Treating the nose, the sinuses and reflux;** speech therapy when nothing is found.

6.2 Common pitfalls

COMMON PITFALLS

- Repeating antibiotics for "catarrh" with no evidence of infection.
- Missing the ACE inhibitor in a patient with chronic cough.
- Operating on the sinuses for postnasal drip alone when the CT and endoscopy are normal.

Written for this app from Oxford Textbook of Otolaryngology ch. 30, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 28. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Maxillary Sinusitis

IMAGING FINDINGS

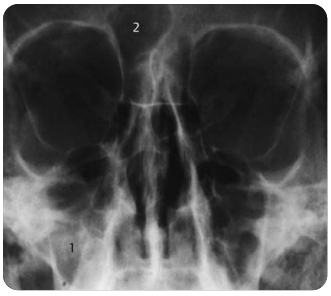
- Air–fluid level, frothy or bubbly secretions
- Mucosal thickening
- Chronic: sclerotic thickened walls (osteitis)

TOP DIFFERENTIALS

- Retention cyst or polyp
- Post-traumatic blood
- Odontogenic source (periapical lucency, oroantral fistula)

CHECKLIST

- Look at the teeth and the ostiomeatal unit
- Contrast CT only for complications



Caldwell view. This 26-year-old patient had pain in the right maxillary region. The right maxillary sinus seems to be opacified in the lower part (1). The lef…

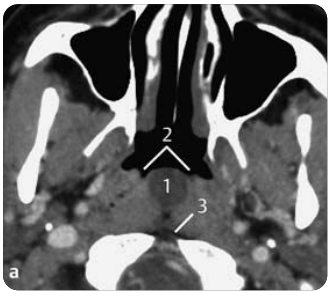


Axial NECT in patient with purulent nasal discharge and headache shows patchy bilateral ethmoid sinus disease, an air-fluid level in the right sphenoid…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 688.

PHARYNX, TONSILS & OESOPHAGUS · CONGENITAL

Thornwaldt Cyst



CT, axial. During endoscopy, a rounded expansion on the posterior wall of the nasopharynx in a patient with sinonasal symptoms was noted. CT…

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

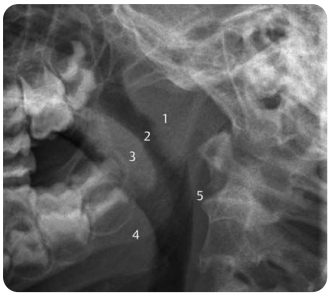
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions



Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2